

Amélie Yang

Engineering Student in Generative AI & Backend Development – IMT Mines Albi

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Summary

Engineering graduate from IMT Mines Albi (Data & Information Systems), with hands-on experience building production-grade generative AI services: backend APIs (FastAPI), RAG pipelines, LLM orchestration and AI agents. Strong interest in healthcare and prevention. Looking for a software/AI engineering position starting September 2026.

Experience

Generative AI Engineering Intern *February 2026 – August 2026 (6 months)*
ALTEN Lab — Grenoble, France

- Designed and built a multimodal RAG service *as-a-service* for industrial document search: containerized FastAPI backend API (Docker), built as a reusable abstraction adaptable to different business domains.
- LLM orchestration, two complementary RAG engines (frameworkless, LangChain/LangGraph); ingestion pipeline (chunking, embeddings), hybrid BM25/semantic search (RRF fusion, PostgreSQL: pgvector + ParadeDB).
- Code quality: benchmarked extraction tools (Docling, PyMuPDF, OCR, VLM), testing, evaluation against academic benchmarks (RAGAS, recall@k, MRR, bootstrap), CI/CD, production deployment on Open WebUI.

R&D Engineering Assistant Intern *April 2025 – August 2025 (4 months)*
Yooz — Aimargues, France

- Built "accountant" AI agents connected to a RAG system indexed on the French General Chart of Accounts, designed for easy integration by third-party clients.
- Developed a web application (Dash) for visualizing and experimenting with AI solutions (semantic search, RAG, LLM), designed to be usable by non-technical business users.
- Clustering and dimensionality reduction (PCA, t-SNE, UMAP) on vector databases (Milvus, ChromaDB) to diagnose and improve semantic search quality in production.

Production Operator Intern *February 2024 – March 2024 (2 months)*
Laboratoires Pierre Fabre — Soual, France

- Strengthened operational skills by managing the packaging line: monitoring, sorting, and feeding processes.

Education

The Catholic University of Korea *September 2025 – December 2025*
AI Major – Exchange Student, Seoul, South Korea
Computer Vision, Data Mining, Machine Learning.

IMT Mines Albi *2023 – 2026*
General Engineering Degree, Albi, France
Optimization, Databases, Numerical Computing, Big Data, Machine Learning.

Lycée Joffre *2021 – 2023*
Preparatory classes MPSI-MP, Montpellier, France

Projects

Handwritten Digit Recognition CNN *2024*
Built a convolutional neural network (CNN) for image classification (MNIST dataset) with PyTorch: convolution/pooling layers, activation functions, backpropagation, optimization (Adam), and regularization (dropout).

Real-Time Platform Waiting-Time Forecasting for SNCF *2025*
Cleaning, processing, and aggregation of flow data. Trained regression and time-series models in Python.

Skills

Programming Languages

Python (FastAPI, Pandas, NumPy, Scikit-learn, PyTorch), SQL, C++

Generative AI / LLM

RAG (as-a-service), LLM orchestration, AI agents, LangChain, LangGraph, LlamaIndex, embeddings

Backend & Infrastructure

FastAPI, integration architectures, Docker, CI/CD, Git

Databases

SQL, BM25 (ParadeDB), vector databases (pgvector, Milvus, ChromaDB)

Machine Learning

Classification, clustering, dimensionality reduction, CNN, time series

Languages

French (native), English (TOEIC 965 pts), Spanish, Japanese

Volunteering

General Secretary – Sports Office, IMT Mines Albi *(2024 – 2025)*

Organized schedules and events. Wrote reports and documentation.

Les Cordées de la Réussite *(2023 – 2024)*

Mentored and guided students in their academic and career orientation.

Certifications

MOOC – Occupational Risk Prevention (INRS) *(2023)*

MOOC – Intellectual Property (INPI) *(2024)*

Interests

Volleyball (2 years competitive)

Music (piano, guitar)

Video games